Thesis: Development of a University Navigator

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Outline

1. Introduction
2. Theoretical Framework
3. Practical Implementation

Introduction

QR Code

https://github.com/ylnt241/eng-project

Scan to read the report, view the presentation or inspect the block diagrams

Thesis Rationale

- Replacement for map.csu.ru
- Need for modern 360° panoramic imaging
- Discontinued old codebase due to:
    - Low cost-effectiveness (technical debt)
    - Need for modern technology stack

Thesis Objectives

- Cross-platform interactive map editor with 3D imagery
- Event and schedule integration, shortest-path routing
- Internal communication: ejabberd server + web client
- Data import tools (parsers) or manual schedule entry

Theoretical Framework

Current Challenges

- No unified secure environment for real-time communication
- Decentralized university information resources
- Low efficiency of notification channels

Solution: Aggregate services into a single interface.

Core Development Principles

- Security, performance, portability, maintainability
- Technology stack selected accordingly
- Excluded: Python, JavaScript, C# (performance / architectural mismatch)
- Goal: gain experience with high-performance systems

Panoramic Content Visualization

- Equirectangular projection for 360° images
- Implemented with `three.js` (virtual sphere / cube mapping)
- Segmented loading (tiling) for 8K+ images

Routing Fundamentals

- Weighted directed graph (Poincaré duality)
- 3D volumes $\rightarrow$ nodes, doors $\rightarrow$ edges
- IndoorGML + `pgRouting` for dynamic subgraph filtering
- Shortest path via Dijkstra's algorithm

Distributed Communication Protocols

- XMPP — open, extensible, XML-based
- `ejabberd` server — fault-tolerant, high concurrency
- Provides:
  - Real-time messaging
  - Multi-user conferences
  - Notifications
  - Isolation of personal social networks

Information Security

- **Shift Left Security** — early vulnerability detection
- Go language — memory safety
- Defenses:
  - ORM (SQL injection prevention)
  - Anubis (bot protection)
  - JWT authorization
  - Middleware, distroless containers
  - Static analyzers + Quality Gates in CI/CD

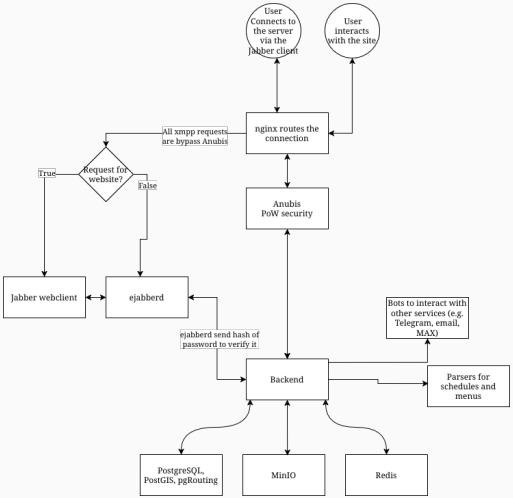
Practical Implementation

QR Code — Architecture Overview

https://github.com/ylnt241/eng-project

Scan to view a high-resolution version of the architecture diagram

General Architecture Overview



Architectural Principles

- **Service Consolidation:** minimize container count
- **Hybrid Monolith:** core logic in a single backend service
- **Stateless Approach:** ready for future clustering

Technology Stack — Core Components

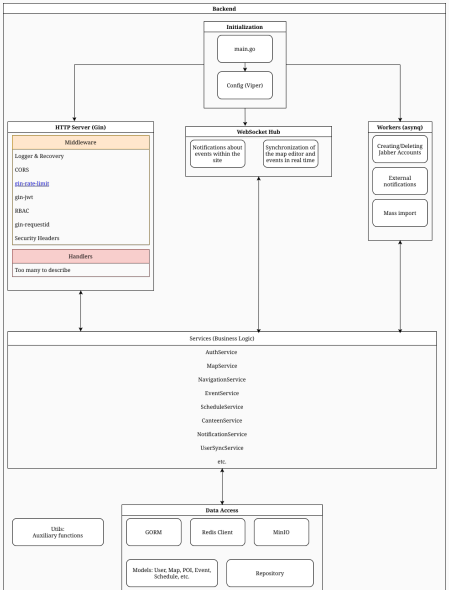
- **Networking:** NGINX + Anubis (bot protection)
- **Storage:** PostgreSQL + PostGIS + pgRouting
- **Communication:** Ejabberd + XMPP client
- **Object Storage & Caching:** MinIO (S3-compatible), Redis
- **External Services:** Python parsers, Go/Python bots

QR Code — Backend Structure

https://github.com/ylnt241/eng-project

Scan to view a high-resolution version of the backend block diagram

Backend Structure



Backend Implementation (Go)

- Multi-layered architecture
- **Initialization:**
  - Viper for configuration
  - Automatic migrations (golang-migrate)
  - Redis & Asynq clients
  - Dependency injection
- **Transport:** Gin framework, WebSocket hub

DevOps and Security

- **CI/CD with Jenkins:**
  - Automated tests on every PR
  - SonarQube static analysis + Quality Gates
  - Mandatory code review
- **Containerization:**
  - Distroless images — minimal attack surface
  - Multi-stage builds

Monitoring and Protection

- Middleware filtering (SQL injection, XSS)
- Anubis + NGINX — block malicious traffic
- Ensures availability under limited resources

Conclusion

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- Modern navigation ecosystem with 360° panoramas
- **Key goals:**
  - Go-based high-performance backend
  - pgRouting + graph model for indoor navigation
  - Shift Left Security, distroless containers
  - Stateless, scalable architecture
- Improves digital interaction for students and staff

The end.

Thank you.